

MIT 805 Project: Part 1

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1 Dataset Provenance and Description

This project uses the NYC Taxi & Limousine Commission’s High Volume For-Hire Vehicle (HVFHV) trip records. The dataset covers trips conducted by Uber, Juno, Via and Lyft and is published through the NYC Open Data Law. Under its FAQs, NYC Open Data states there are no restrictions on the use of Open Data. However, the TLC page does include a standard public accuracy disclaimer but this does not limit academic use. Thus, the dataset is appropriate for the project (New York City Taxi and Limousine Commission (n.d.), NYC Open Data (n.d.)).

2 Dataset Details

The raw dataset consists of 73 monthly Apache Parquet files (30.21 GB) covering February 2019 to May 2025. After cleaning and selecting 11 relevant columns, the working dataset contains 1,283,180,517 records across 73 files (25.22 GB). A 5% sample of 64,159,022 records was used for exploratory analysis to reduce computational cost while maintaining coverage across the study period. Key characteristics include the large scale nature of the data, schema changes across years, and its focus on app-dispatched high volume for-hire vehicle trips. For a detailed breakdown please refer to Table 1 in Appendix A.

3 Data Preparation and Quality Concerns

Preparation: Each monthly file was reduced to 11 relevant variables and cleaned by removing records with missing or impossible distance, duration, or fare values. A 5% random sample was then taken from each month for exploratory analysis, reducing computational requirements while maintaining coverage across 2019–2025.

Quality concerns: The main limitations were **schema changes** across years, **extreme but potentially valid outliers**, and the fact that HVFHV data only represents app-based high volume for-hire vehicles. The 5% sample used for EDA also introduces a computational trade-off between computational efficiency and completeness. Further details in Appendix B

4 Exploratory Data Analysis Summary

The EDA shows a clear **COVID-19 impact**, with monthly trips falling drastically in April 2020 before recovering to previous levels. Demand is lowest around **3–5am** and highest during the **5–7pm evening peak**, with Friday and Saturday generally the busiest days.

Trip distance, duration, and base fare are **right-skewed**, with a small number of extreme trips. Distance and base fare are strongly positively correlated ($r = 0.85$), while only around **6.2% of trips** involved a shared-ride request. Refer to Appendix B for further results and supporting visuals.

5 Business Value and Limitations

Business value: The dataset provides value for transport-policy, labour and resilience analysis. Between January and May 2025,, **31–35%** of trips each month included a fee for. This represented by 7.24 million of 21.09 million trips (**34.3%**) in May. This brings business value because it provides regulators with information to assess how congestion pricing affects the ride-hail market and the additional cost faced by passengers. Essentially this would help regulators in judging whether policies are fairly distributed across cities.

The data can also support labour-equity analysis. The average driver pay increased from \$16.88 per trip before December 2023 to \$20.17 afterward, providing a basis for monitoring labour-market outcomes over time. However, this before-and-after comparison does not establish that the pay-rule change caused the increase. The data also highlights the significant impact of COVID-19 on monthly trip volumes, showing a drastic decrease in April 2020. Trip volumes returned toward normal levels in 2022. These findings provide valuable insights for public health and urban planning by showing how transport demand responds to major disruptions. This can help cities improve the current transport systems and prepare for future disruptions.

Limitations: One limitation of this dataset is the evidence of schema drifting, where fields such

as `airport_fee`, `cbd_congestion_fee`, and `wav_match_flag` change data types over the study period. For example, `wav_match_flag` was stored as an integer in 2019 but as a string in 2025, which creates some inconsistencies that need to be handled when combining the monthly files.

Another limitation is the overlapping time periods of major events. COVID-19, the 2023 driver-pay rule, and congestion pricing in 2025 all occur within the dataset. Thus, the analysis is mainly looking at what changed before and after these events instead of proving that the change was policy driven. This makes it difficult to attribute any observed changes to one policy.

6 Analysing the 7 Vs of Big Data

Volume: The raw dataset is approximately 30 GB across 73 monthly Parquet files, with the cleaned working dataset containing over 1.28 billion records. Processing this on a single machine would be expensive, making distributed processing with tools like PySpark more appropriate.

Velocity: Trip volumes differ over time, with some months containing significantly more records than others, with a sharp drop during the COVID-19 period and recovery by May 2025. If the data was processed continuously as subsequent records become available, the system would need to handle changing ingestion volumes efficiently.

Variety: The raw dataset contains more than 20 columns depending on the year and includes several types of data. These include numerical fields such as fares and distances, timestamps for pickup and drop-off times, categorical flags such as `shared_request_flag` and `wav_request_flag` and identifiers such as location and licence numbers.

Value: The dataset provides value for various transport related use cases. It has the ability to analyse congestion pricing policies and driver pay, as well as sufficient data to study the decline and recovery of the industry during COVID-19. This understanding can assist with transport planning, policy reviews and analysis of industry resilience.

Veracity: The dataset has some data quality and consistency issues across the years. Fields such as `airport_fee`, `cbd_congestion_fee` and `wav_match_flag` change in availability or format over time. Extreme values are also present after cleaning. These issues need to be checked and handled carefully to ensure reliable analysis.

Variability: The relevance of some fields change over time as the operating environment changes. For example, `cbd_congestion_fee` only becomes relevant in the later data when congestion pricing is introduced.

Volatility: Changes in fare structures, driver pay rules and congestion pricing affects the business and regulatory context over the study period. As a result, relationships identified during one period may not remain valid as new data is generated. Analyses and models therefore need to be reviewed over time to ensure that their conclusions remain relevant.

7 References

New York City Taxi and Limousine Commission (n.d.) *TLC Trip Record Data — High Volume For-Hire Vehicle (HVFHV) trip records* URL: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page> (visited on 08/27/2026).

NYC Open Data (n.d.) *Frequently Asked Questions* URL: <https://opendata.cityofnewyork.us/faq/> (visited on 08/27/2026).

Pillay, T and Smit, R (2026) *MIT 805 Group Project* https://github.com/Thigan/MIT_805_Group_Project GitHub repository, accessed 3 September 2026.

Appendix

A Data Characteristics and Details

Table 1: Overview of the HVFHV dataset used in this project.

Field	Detail
Source	NYC Taxi & Limousine Commission (TLC) official trip record data
Dataset	High Volume For-Hire Vehicle (HVFHV) trip records
Format	Apache Parquet, with one file per calendar month
Variables	Raw: 20–24 columns depending on year. Working: 11 selected variables
Data period	February 2019–May 2025, covering 73 monthly files
Working records	1,283,180,517 records after data preparation
EDA sample	64,159,022 records, representing a 5% random sample drawn across the monthly working files
Raw dataset size	30.21 GB across 73 files
Working dataset size	25.22 GB across 73 files
Processing dataset size	Not yet finalised. This will represent the portion of the working dataset read and transformed by the PySpark pipeline in Part 2 (target ≥ 3 GB).

A.1 Data Quality Summary

Table 2: Summary of key data quality observations from the EDA sample.

Quality issue	Observation
Missing values	No missing values were identified in the 11 variables of the cleaned EDA sample.
Extreme trip distance	Maximum observed trip distance was 612.44 miles.
Extreme trip duration	Maximum observed trip duration was 71,606 seconds (approximately 19.9 hours).
Extreme base fare	Extreme base passenger fare values remained after cleaning.
Driver pay	Negative driver-pay values remained.
Schema consistency	Some fields changed in availability or representation across different years.

B Exploratory Data Analysis

The following figures provide supporting visualisations for the exploratory data analysis.

B.1 Demand Patterns

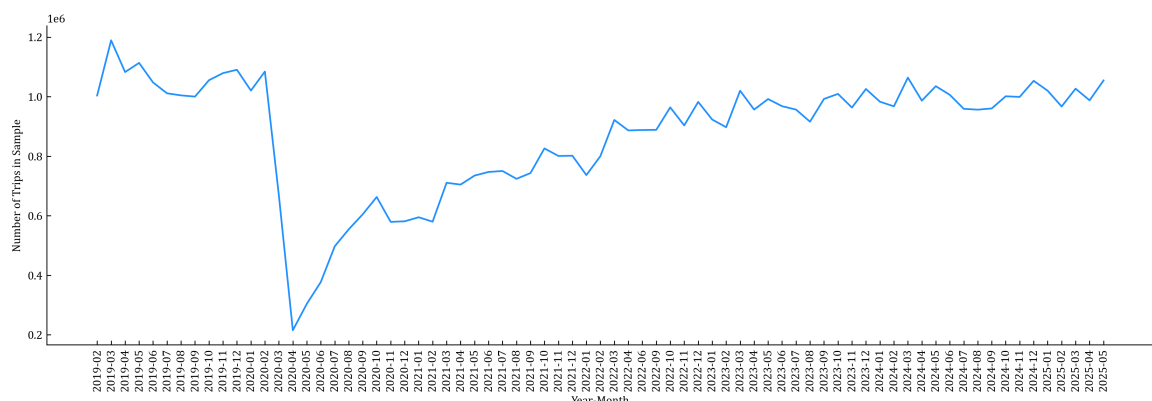


Figure 1: Number of sampled HVFHV trips per month from February 2019 to May 2025.

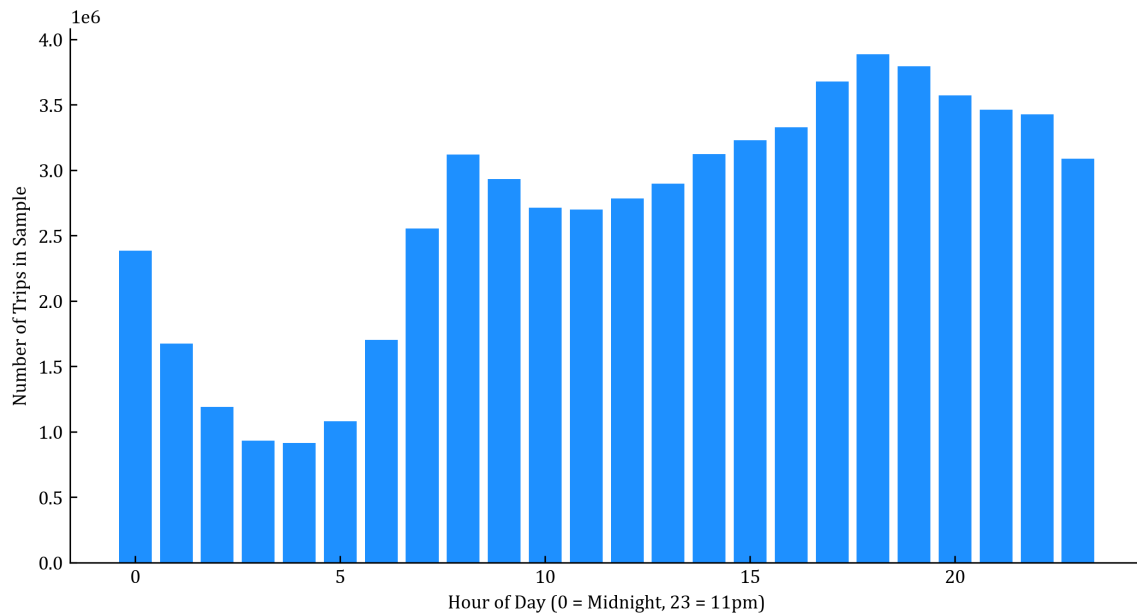


Figure 2: Distribution of sampled trips by hour of day.

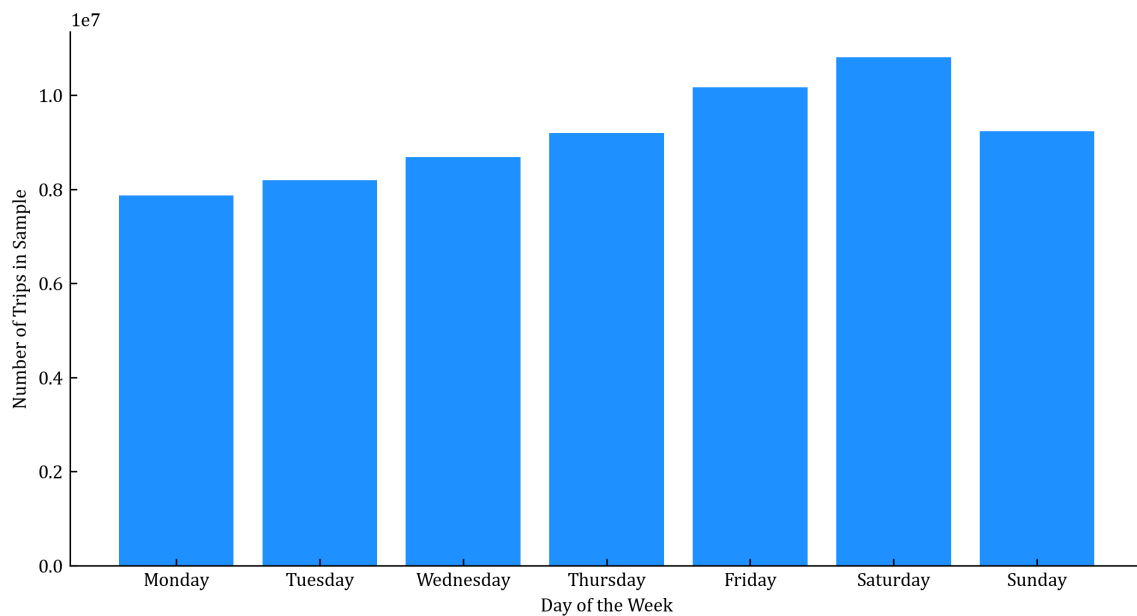


Figure 3: Distribution of sampled trips by day of the week.

Key insight: Trip demand dropped sharply during the COVID-19 period, falling from approximately 20.4 million trips in January 2020 to 4.3 million in April 2020 before gradually recovering.. Looking at the sampled data, demand is lowest between 3–5am, and highest between 5–7pm. Friday and Saturday are generally the busiest days.

B.2 Trip and Fare Characteristics

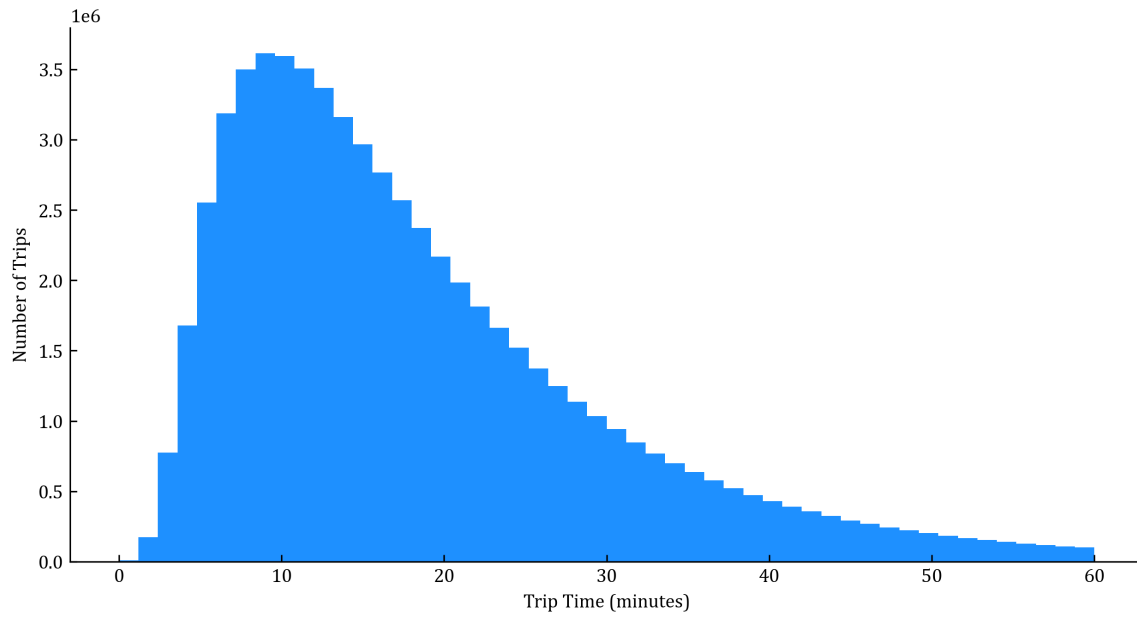


Figure 4: Distribution of trip duration in the sampled HVFHV data.

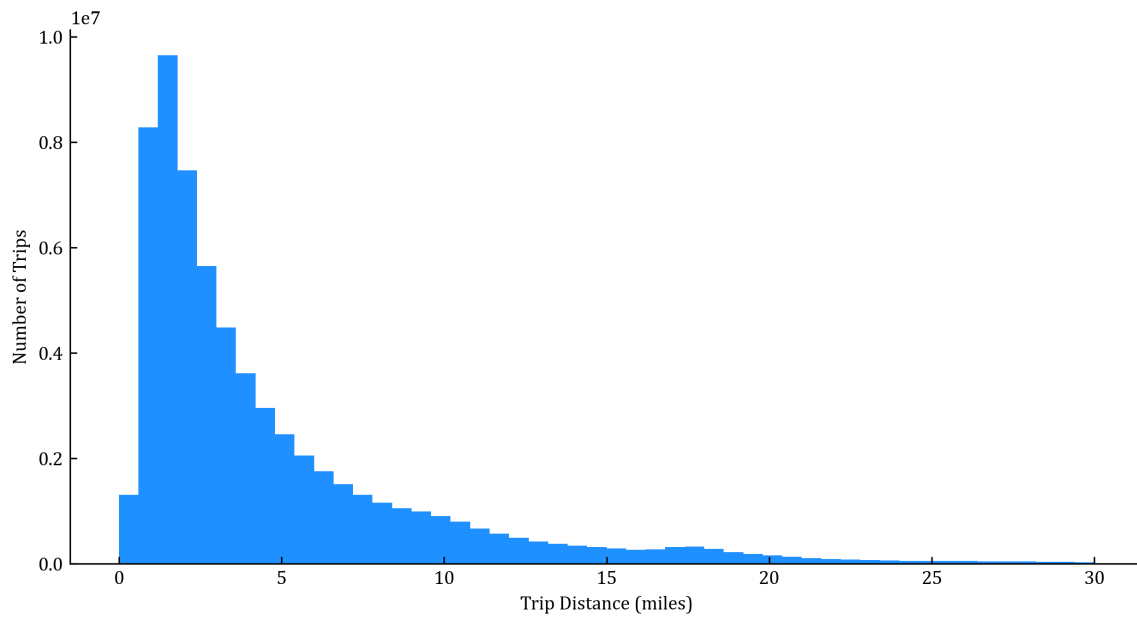


Figure 5: Distribution of trip distance in the sampled HVFHV data.

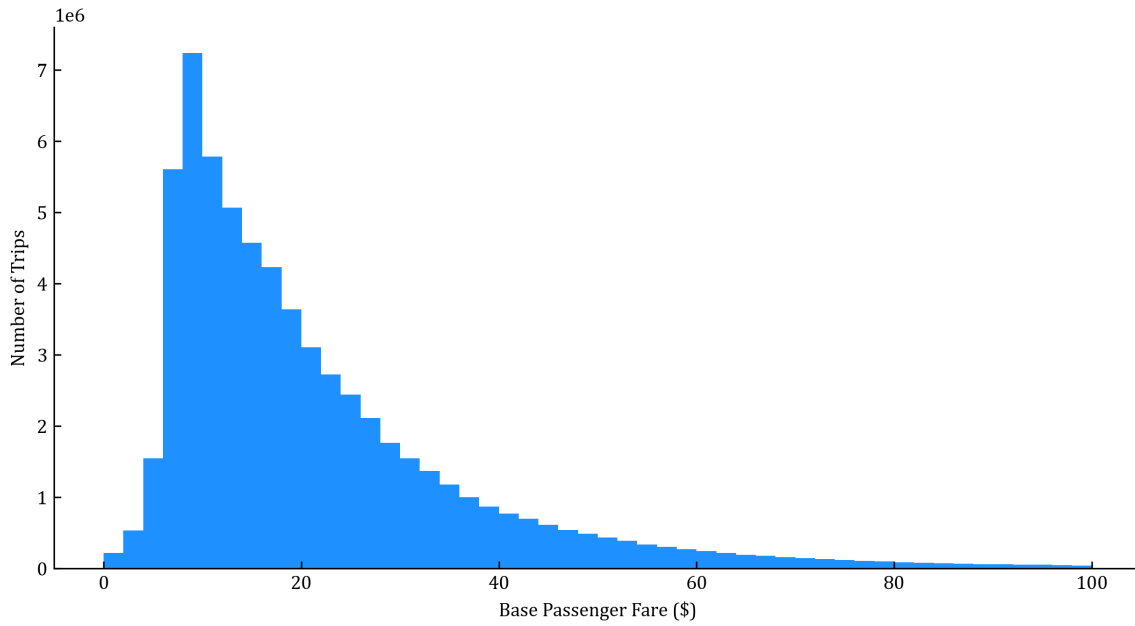


Figure 6: Distribution of base passenger fares in the sampled HVFHV data.

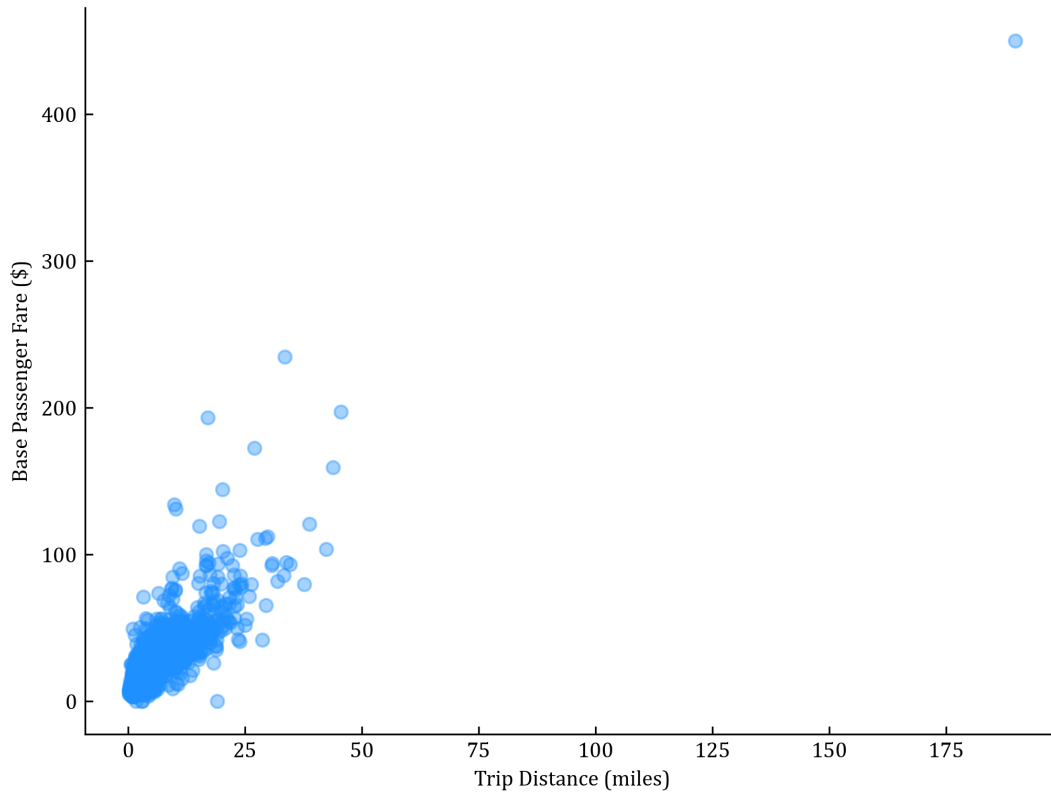


Figure 7: Relationship between trip distance and base passenger fare for a random sample of 2,000 trips.

Key insight: Trip duration, distance and base passenger fare are strongly right skewed. 404,732 of the 64.16 million sampled trips exceeded 30 miles, 1,110,460 exceeded 60 minutes and 653,602 had base passenger fares above \$100. Trip distance and base passenger fare shows a strong positive correlation ($r = 0.851$).

B.3 Shared Rides and Pickup Demand

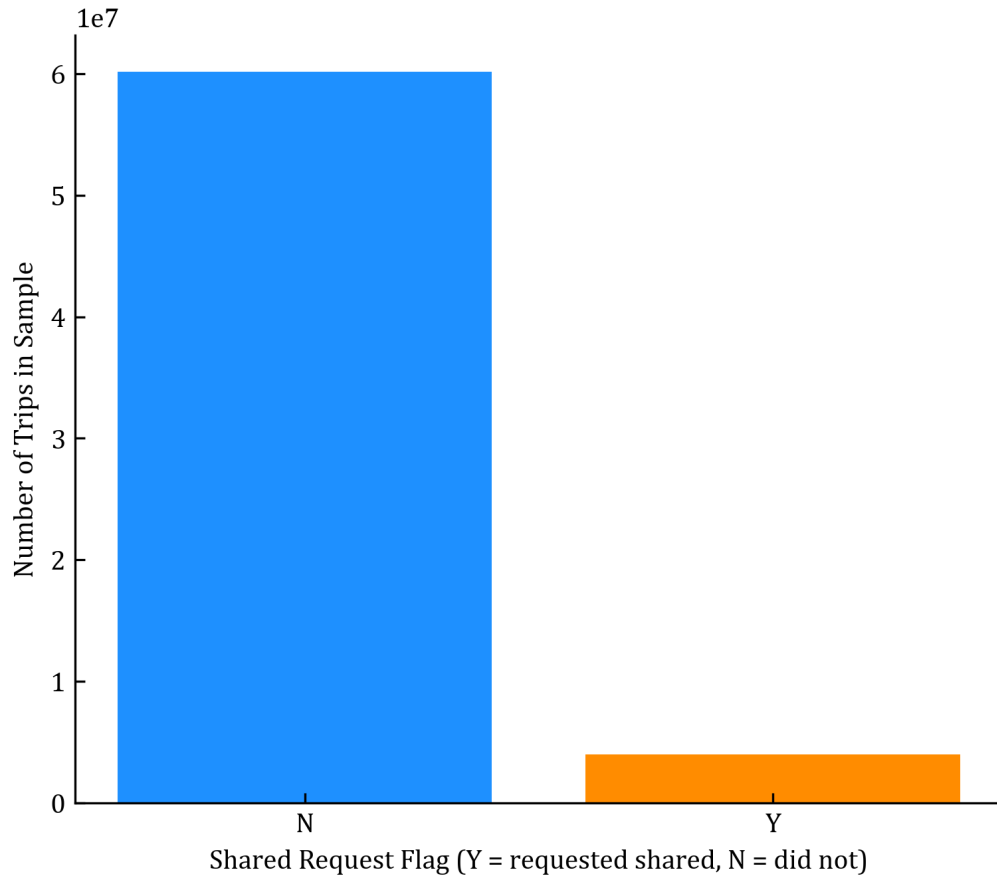


Figure 8: Comparison of requested shared rides and solo rides in the sampled data.

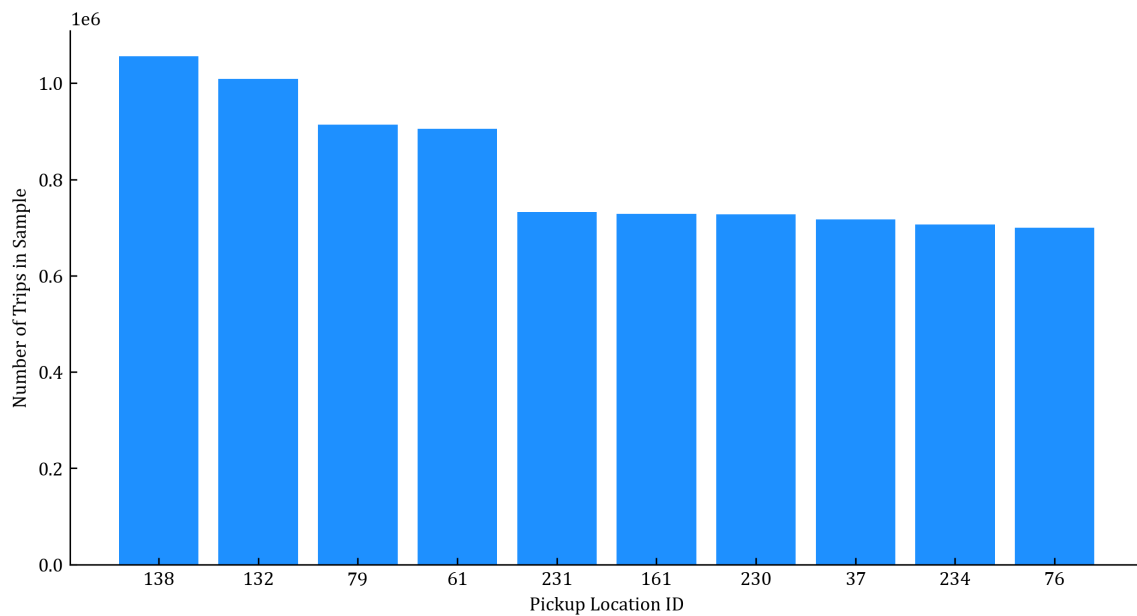


Figure 9: Top 10 busiest pickup zones in the sampled data by TLC Location ID.

Key insight: Shared rides are uncommon, representing roughly 6.2% of the sampled trips. Pickup activity is concentrated around a small number of TLC Location IDs, indicating that HVFHV demand is not evenly distributed across pickup locations.

C Supporting Business Analysis

The following results provide additional support for the business value analysis presented in the main report.

C.1 Congestion Pricing

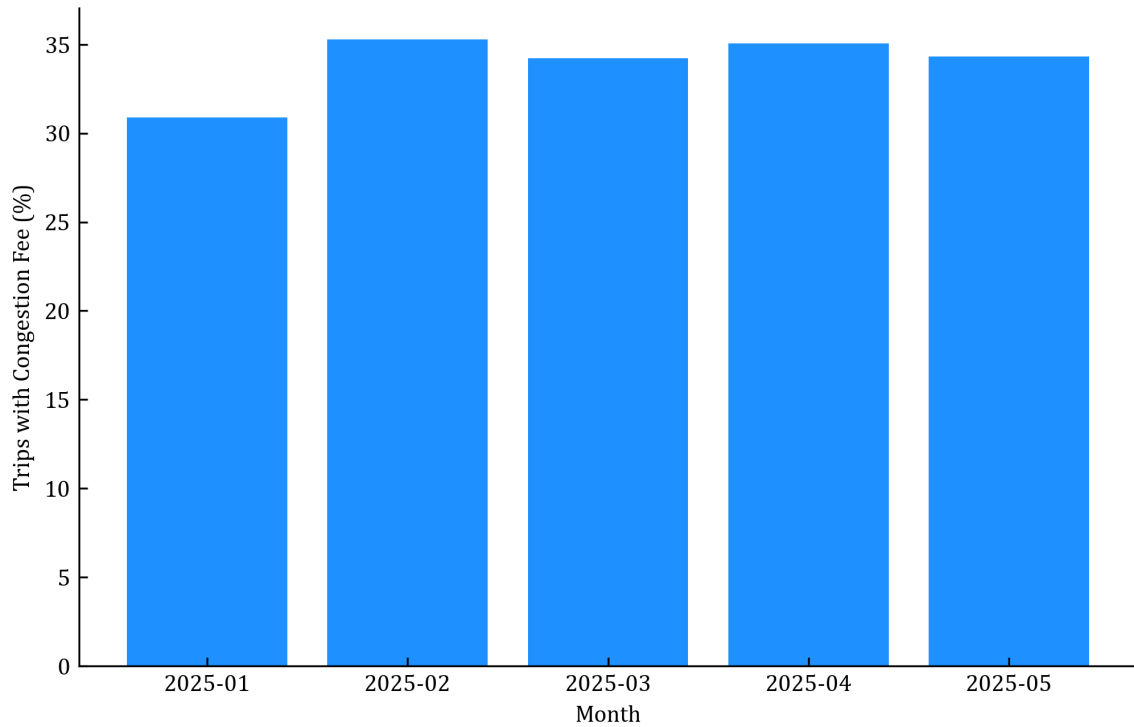


Figure 10: Percentage of HVFHV trips recording a congestion fee from January to May 2025.

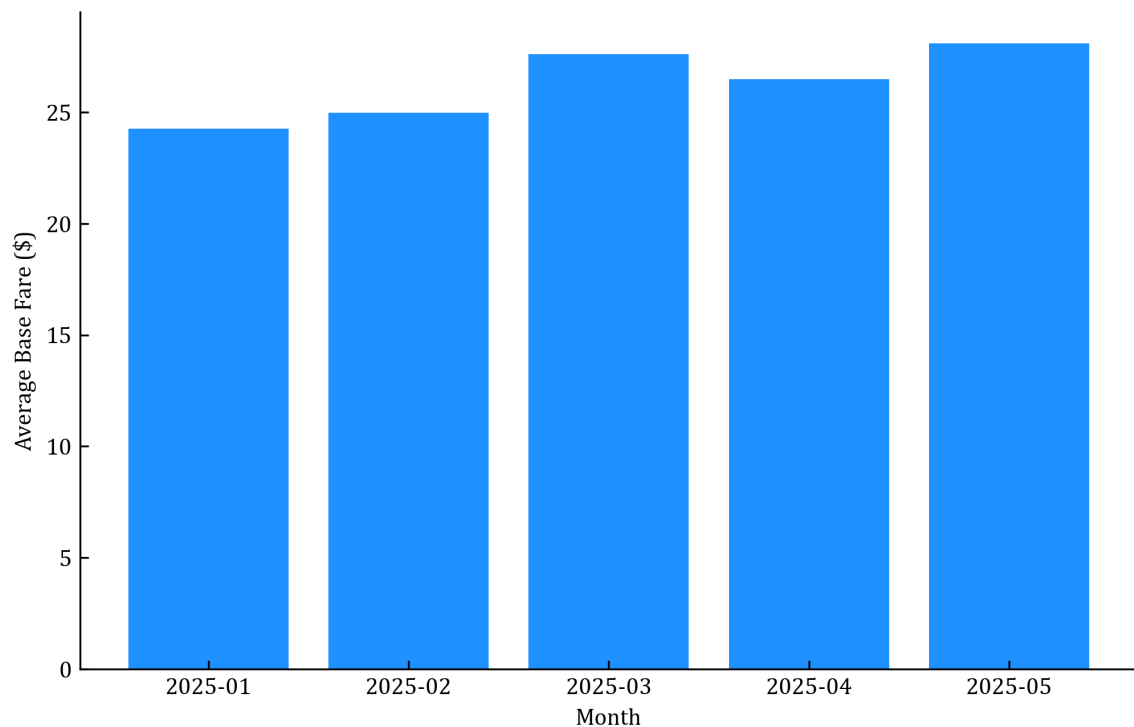


Figure 11: Average base passenger fare from January to May 2025.

Key insight: 31–35% of HVFHV trips recorded a congestion fee from January to May 2025. In May, 7.24 million of 21.09 million trips (34.3%) recorded a fee. Average base passenger fares ranged from \$24.27 to \$28.10. These results simply describe the observed patterns but does not

establish a correlation to changes in passenger fares.

C.2 Driver Pay

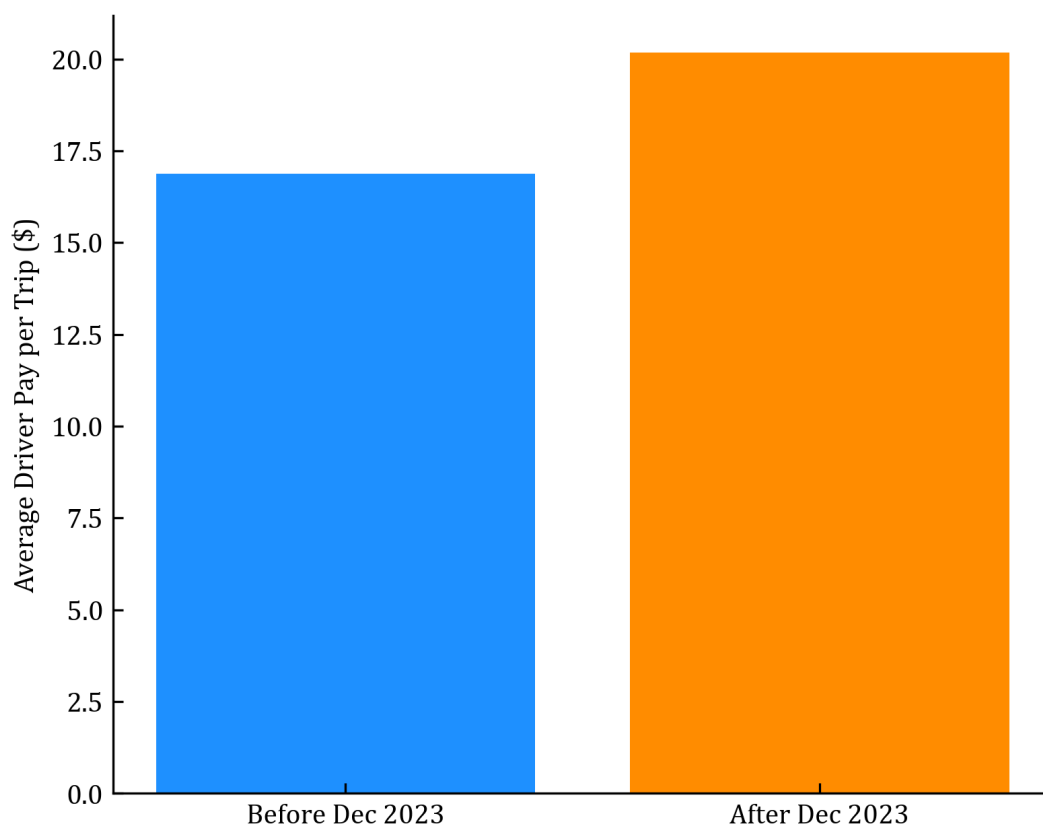


Figure 12: Average recorded driver pay per trip before and after December 2023.

Key insight: Average recorded driver pay increased from \$16.88 per trip before December 2023 to \$20.17 afterward. This can be used to monitor driver remuneration. However, other changes observed over the same time frame cannot be attributed to the pay rule change alone.